

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 08

Lab Report Name : Inter-VLAN Routing in Cisco Packet Tracer

Course Title : Computer Networks Lab

Course Code : ICT-3202

Submitted By	Submitted To
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Date of Performance :

Date of Submission :

Experiment No: 08

Experiment Name: Inter-VLAN Routing in Cisco Packet Tracer

Introduction:

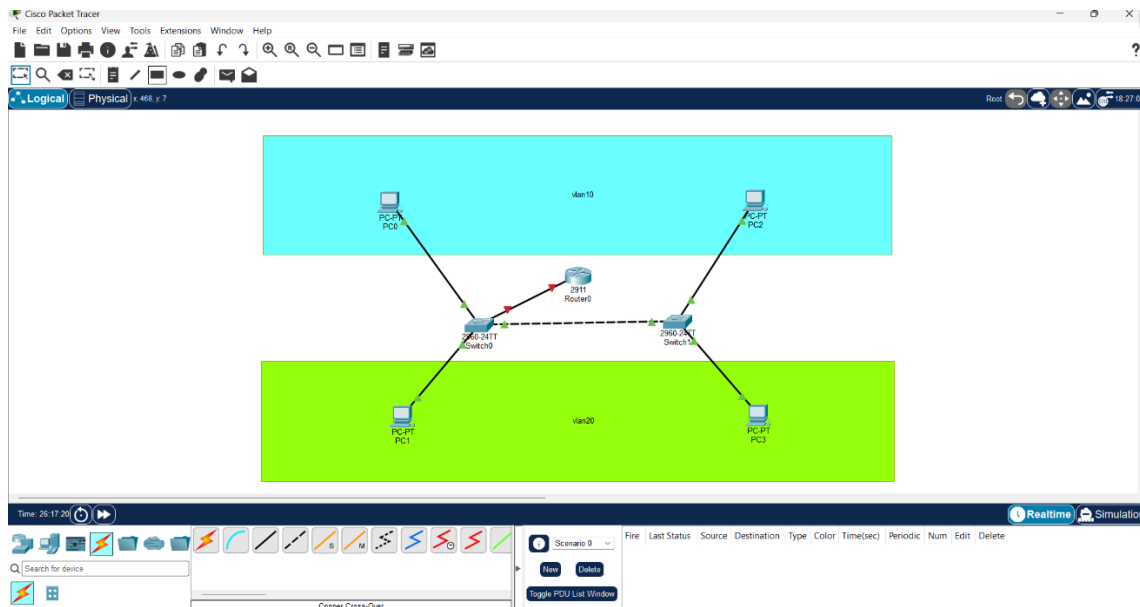
This lab demonstrates the implementation of virtual local area networks (VLANs) across multiple switches and the configuration of inter-VLAN routing using a single physical router interface (Router-on-a-Stick methodology). The topology isolates traffic between two departments—Sales (VLAN 10) and HR (VLAN 20)—and routes traffic between them using IEEE 802.1Q sub-interface encapsulation.

Network Topology & Addressing Scheme:

The network consists of two Cisco Catalyst 2960 switches, one Cisco 2911 router, and four host PCs partitioned into two logical VLAN networks.

Device	Interface	IP Address	Subnet Mask	Default Gateway	Assigned VLAN
PC0	FastEthernet 0	192.168.10.x	255.255.255.0	192.168.10.1	VLAN 10 (Sales)
PC1	FastEthernet 0	192.168.20.x	255.255.255.0	192.168.20.1	VLAN 20 (HR)
PC2	FastEthernet 0	192.168.10.x	255.255.255.0	192.168.10.1	VLAN 10 (Sales)
PC3	FastEthernet 0	192.168.20.11	255.255.255.0	192.168.20.1	VLAN 20 (HR)
Router0	GigabitEthernet 0/0.10	192.168.10.1	255.255.255.0	N/A	VLAN 10 Gateway
Router0	GigabitEthernet 0/0.20	192.168.20.1	255.255.255.0	N/A	VLAN 20 Gateway

Device Configurations:



1. Switch0 Configuration:

Switch0 defines both VLANs, assigns access ports to local hosts, and configures trunk ports toward Router0 and Switch1.

```
Switch>
Switch>
Switch>cc
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan10
^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
Switch(config)#
```

2. Switch1 Configuration:

Switch1 replicates the local VLAN database and maps local host ports while establishing the trunk connection back to Switch0.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

3. Router0 Configuration :

The physical interface GigabitEthernet 0/0 is enabled, and logical sub-interfaces are created to handle tagging for VLAN 10 and VLAN 20.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Verification & Testing:

Connectivity was validated from **PC0** (VLAN 10) targeting **PC3** (192.168.20.11, VLAN 20):

- **Initial Test:** The first echo request timed out due to initial Address Resolution Protocol (ARP) frame processing across the sub-interfaces, resulting in a 25% packet loss (3/4 received).
- **Secondary Test:** Subsequent ICMP Echo requests yielded a **100% success rate (0% packet loss)** with an average round-trip time of <1 ms.

```
PC0

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.11: bytes=32 time=32ms TTL=127
Reply from 192.168.20.11: bytes=32 time=1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 32ms, Average = 11ms

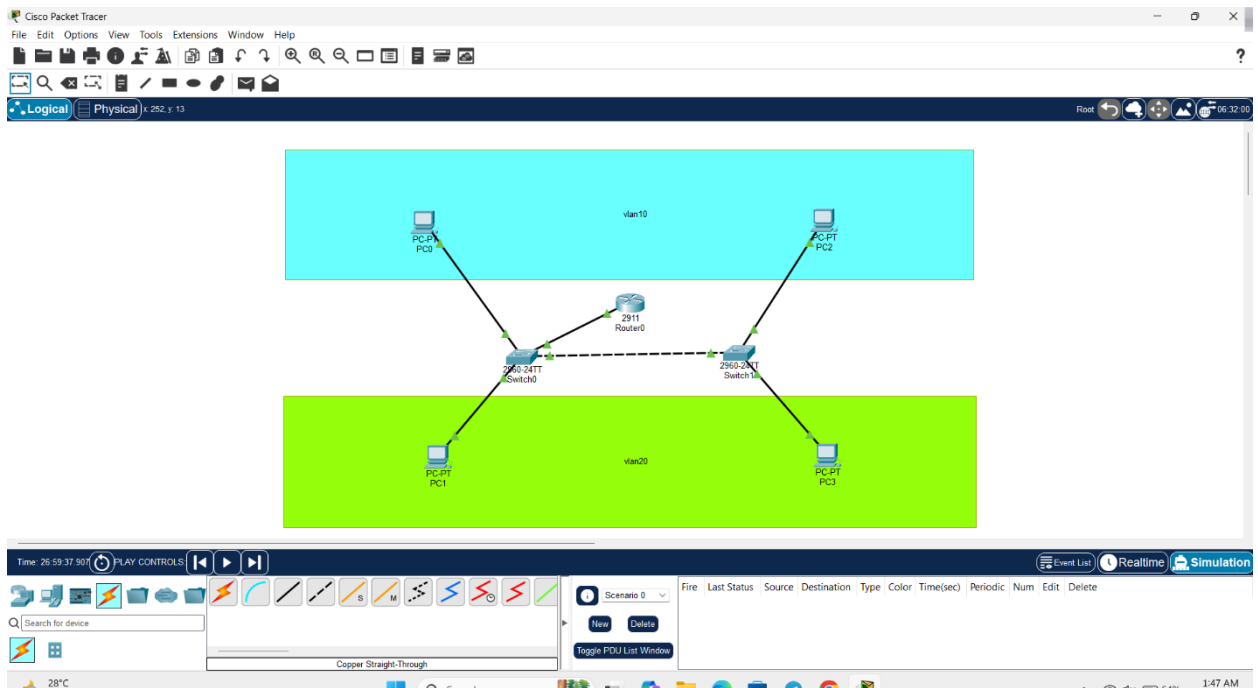
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```



Conclusion

The Router-on-a-Stick deployment was completed successfully. Virtual network segmentation was maintained at Layer 2 across multiple switches while facilitating full inter-VLAN routing at Layer 3 through 802.1Q sub-interface encapsulation. All interface link states are active and green.